

LAB 3 – ARCHITECTURAL PATTERN JUSTIFICATION

Problem Statement #39: Second-Hand Product Escrow Marketplace

Architecture Selection: Microservices Architecture

The marketplace is decomposed into independently deployable services: Buyer UI, Order/Campaign Manager, Escrow Payment Service, Inspection and Verification Service, Dispute Resolution Service, Notification Service, and a database. The Escrow Payment Service communicates with an external payment gateway. Services interact through authenticated REST/HTTPS APIs and asynchronous messaging where appropriate.

1. Reason 1 – Secure escrow and dual verification (FR-001, FR-004)

The central rule is that funds remain locked until the required verification is complete: the inspection agent certifies the product condition and the buyer provides sign-off. The Escrow Payment Service exposes a narrow fund-release interface that can be invoked only after both conditions are satisfied. Dispute Resolution can freeze escrow immediately when a ticket is raised. Centralizing release, refund, and freeze operations makes them easier to audit and reduces the risk of premature payment.

2. Reason 2 – Independent scaling and change (FR-005, NFR-002)

Marketplace browsing and order activity may experience higher traffic than inspection or dispute handling. Microservices allow the UI-facing catalogue/order components to scale independently from inspection, escrow, and dispute services. Rules such as inspection workflows or dispute handling can be updated without redeploying every component. Asynchronous notifications prevent email or in-app delivery from blocking checkout or verification.

3. Security Advantage (NFR-001)

The Escrow Payment Service alone handles payment tokens and communicates with the external payment gateway over HTTPS. Raw card details should not be stored by the marketplace; only tokens and transaction references are retained where required. The Dispute Resolution Service can place an immediate hold on funds and block withdrawals during investigation. Role-based access ensures that only authorized inspection agents can submit inspection results and only the buyer can provide buyer sign-off.

4. Performance Benefit (NFR-002)

Services can scale horizontally according to demand. Order and payment processing can receive additional instances during peak browsing and purchase periods, while lower-volume inspection and dispute services remain independently sized. A message queue handles notifications asynchronously. The precise response-time and concurrency targets should be defined in the requirements table and validated with load tests, since the provided problem statement does not specify numeric targets.

5. Trade-offs Accepted

Microservices introduce network latency, operational complexity, distributed failure modes, and consistency challenges. These can be mitigated through idempotent escrow operations, authenticated service-to-service calls, durable audit records, retry policies, and explicit order-state transitions. For a marketplace that holds buyer funds and must prevent premature release, the isolation and auditability benefits justify considering these trade-offs.

ARCHITECTURE COMPONENTS AND INTERFACES

Component	Responsibility	Key interfaces
Buyer UI	Browse listings, place orders, confirm receipt/sign-off, raise disputes.	REST/HTTPS to Order Service
Order Service	Maintain order state and coordinate inspection workflow.	IOrder; REST/HTTPS
Escrow Payment Service	Lock funds, release/refund funds, freeze escrow, record transactions.	IEscrow; IGateway over HTTPS
Inspection & Verification Service	Record inspector certification and buyer sign-off status.	IInspection; authenticated API
Dispute Resolution Service	Open tickets, freeze escrow, manage investigation status.	IDispute; event/API
Notification Service	Send order, inspection, payment, and dispute updates.	Message queue
Database / Audit Store	Persist order state, verification records, tokens/references, and audit trail.	SQL/JDBC; access-controlled

Important: The architecture above is a proposed justification based on Problem Statement #39. The supplied Lab 3 sample is for Problem #40, so its exact FR/NFR numbering and numeric performance targets should not be copied unless they are defined in the Problem #39 requirements.